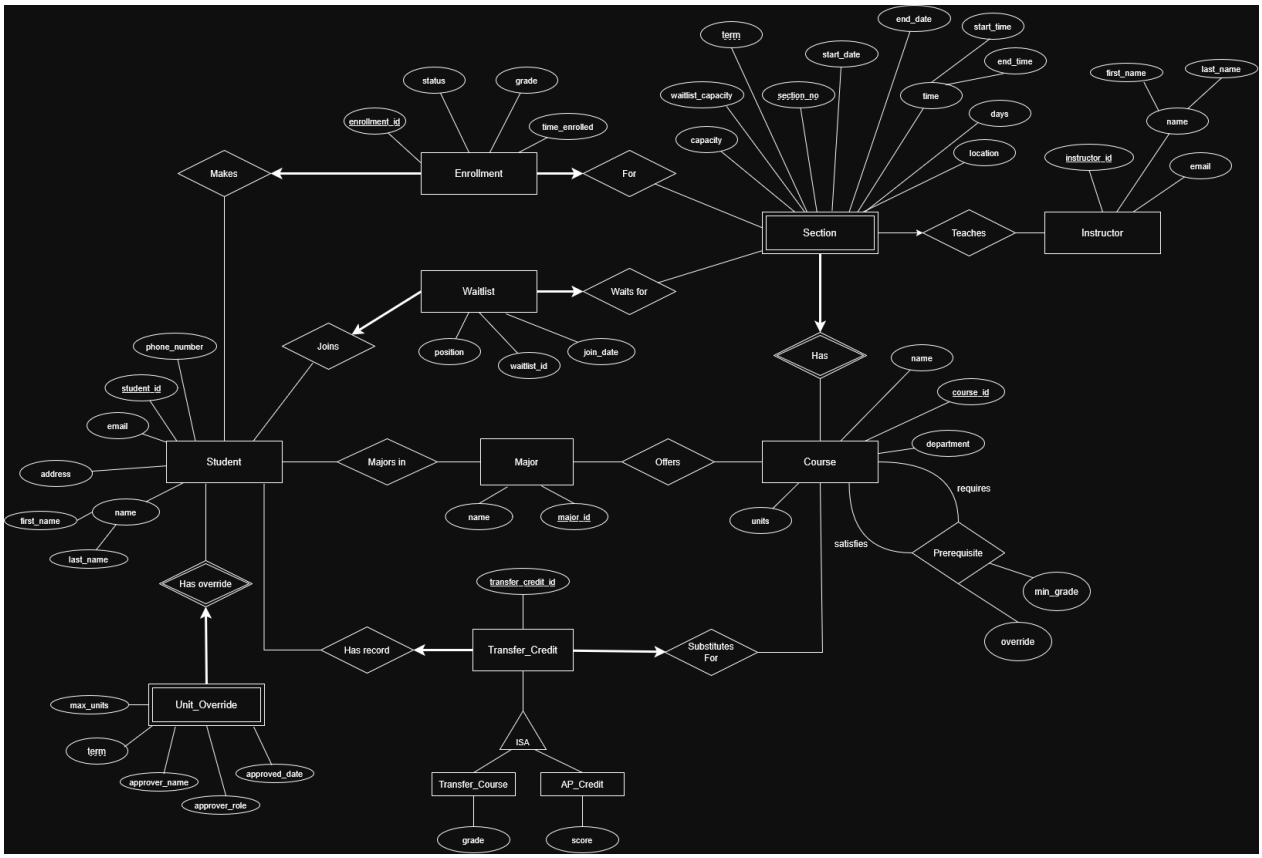
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Project Enrollment
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ER Diagram



Relational Mapping

Relation	ON DELETE
Student(**student_id**, first_name, last_name, address, phone_number, email)	
Enrollment(**enrollment_id**, student_id, course_id, term, section_no, status, grade, time_enrolled)	FOREIGN KEY (student_id) REFERENCES Student(student_id) ON DELETE CASCADE FOREIGN KEY (course_id, term, section_no) REFERENCES Section(course_id, term, section_no) ON DELETE CASCADE

Waitlist(**waitlist_id**, student_id, course_id, term, section_no, position, join_date)	FOREIGN KEY (student_id) REFERENCES Student (student_id) ON DELETE CASCADE FOREIGN KEY (course_id, term, section_no) REFERENCES Section(course_id, term,section_no) ON DELETE CASCADE
Course(**course_id**, name, units, department)	
Section(**course_id, term, section_no,** start_date, end_date, capacity, waitlist_capacity, days, start_time, end_time, location, instructor_id)	FOREIGN KEY (course_id) REFERENCES Course(course_id) ON DELETE CASCADE FOREIGN KEY (instructor_id) REFERENCES Instructor (instructor_id) ON DELETE SET NULL
Instructor(**instructor_id**, first_name, last_name, email)	
Unit_Override(**student_id**, **term**, max_units, approver_name, approver_role, approved_date)	FOREIGN KEY (student_id) REFERENCES Student(student_id) ON DELETE CASCADE
Major(**major_id**, name)	
Transfer_Course(**transfer_credit_id**, grade, student_id, course_id)	FOREIGN KEY (student_id) REFERENCES Student(student_id) ON DELETE CASCADE FOREIGN KEY (course_id) REFERENCES Course(course_id) ON DELETE SET NULL
AP_Credit(**transfer_credit_id**, score, student_id, course_id)	FOREIGN KEY (student_id) REFERENCES Student(student_id) ON DELETE CASCADE FOREIGN KEY (course_id) REFERENCES Course(course_id) ON DELETE SET NULL

Major_Course_Offering(major_id, course_id)	FOREIGN KEY (major_id) REFERENCES Major(major_id) ON DELETE CASCADE FOREIGN KEY (course_id) REFERENCES Course(course_id) ON DELETE CASCADE
Student_Major(student_id, major_id)	FOREIGN KEY (student_id) REFERENCES Student(student_id) ON DELETE CASCADE FOREIGN KEY (major_id) REFERENCES Major(major_id) ON DELETE RESTRICT
Course_Prerequisite(course_id, prereq_course_id, min_grade, override)	FOREIGN KEY (course_id) REFERENCES Course(course_id) ON DELETE CASCADE FOREIGN KEY (prereq_course_id) REFERENCES Course(course_id) ON DELETE RESTRICT

FD's
1. (student_id) → (first_name, last_name, address, phone, email)
2. (course_id) → (course_name, units, department)
3. (course_id, section_no, term_name) → (capacity, waitlist_capacity, start_date, end_date, days, start_time, end_time, location, instructor_id)
*section overlap taught by instructors should be a constraint.
4.
(student_id, section_id) → (status, grade, time_enrolled)
(enrollment id) → (student id, section id, status, grade, time enrolled)
5.
(instructor_id) → (first_name, last_name)
6.
(course_id, prerequisite_course_id) → (minimum_grade, override)
*ignored substitutes for now
7. (student_id, section_id, term_name) → (join_date, position)
8. (student_id, term_name) → (max_units, override)
9.
*Schedule constraint to handle overlap.
Additional Primary Keys:
1. (student id, major_id)
2. (course_id, major_id)

View and Trigger Justifications

Triggers are needed because these rules have to count, add up or search many rows, which a CHECK, UNIQUE or FOREIGN KEY cannot do.

trg_enrollment_capacity: counts enrolled students in a section and blocks the enrollment if it would go over capacity, locking the section row (FOR UPDATE) so two students can't take the last seat at once.

trg_waitlist_capacity: counts students on a section's waitlist and blocks joining if it would go over waitlist_capacity.

trg_enrollment_units: adds up a student's enrolled units for the term and blocks the enrollment if it goes over 18, or over their Unit_Override limit.

trg_enrollment_prereqs: checks Enrollment, Transfer_Course and AP_Credit to make sure the student passed each prerequisite with at least min_grade.

trg_enrollment_schedule: compares the new section's days and times with the student's other sections that term and blocks any overlap.

trg_section_instructor: compares a section's days and times with the instructor's other sections that term and blocks any overlap.

trg_waitlist_not_enrolled: stops a student from joining the waitlist of a section they are already enrolled in.

trg_enrollment_leave_waitlist: removes a student from a section's waitlist when they enroll in it, and moves everyone behind them up.

trg_waitlist_promote: when a student drops, enrolls the next eligible waitlisted student in the same transaction, which is our signature transaction.

Section_Seat_Status (view): shows each section's capacity, enrolled count, open seats and waitlist size, calculated fresh on every query so the numbers can't get out of sync like stored counts could.